TERM PROJECT / TECHNICAL BLUEPRINT

UF Policy RAG Chatbot

Deliverable 2 · November 11, 2026

1. Project objective and work completed

I started the UF Policy RAG Chatbot from scratch for this course. Students often ask questions in everyday language, while official policies use formal terms and contain conditions and exceptions. The chatbot finds relevant passages, generates a concise answer, and lets users inspect citations and official sources.

Component built for this project	Current implementation
Policy collection	Web scraper and collection notebook produced 1,887 text chunks across 189 policy titles.
Retrieval	Pretrained `BAAI/bge-base-en-v1.5` and the persistent ChromaDB `policies` collection.
Answer generation	`gpt-oss-20b` through the UF AI gateway.
Streamlit chat	Inline citations, retrieved excerpts, official links, and adjustable passage count.
Exploratory evaluation	51 questions compared using BLEU, ROUGE, and METEOR.

Current evaluation completed. The 51-question workflow measures answer-text similarity. The four retrieval comparisons, Recall@5, mean reciprocal rank (MRR), and systematic citation-support scoring are planned for the next deliverable.

Research question. Does fine-tuning BGE-small on synthetic student-style questions improve policy retrieval over BM25 and pretrained BGE-small, especially for questions that share few words with the policy? I expect fine-tuning to improve Recall@5 over the same pretrained checkpoint. The hybrid will test whether combining keyword and dense retrieval adds value.

2. Data and collection design

The corpus comes from the public UF Regulation and Policy Hub. The snapshot includes a policy JSON file, a source-link JSON file, and an Excel workbook. Its 189 titles describe the collected snapshot. The next deliverable will audit coverage and record unavailable documents.

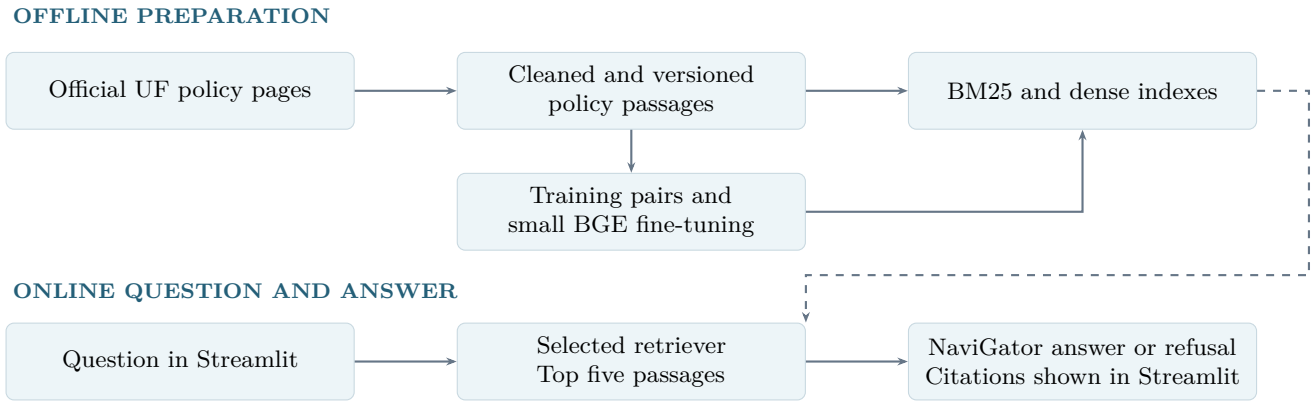
Next deliverable. Audit and freeze a dated corpus. Retain source URLs, titles, headings, stable section IDs, and passage text. Remove repeated boilerplate and duplicates while preserving conditions and exceptions. Hold section boundaries and passage preparation constant across retrieval systems.

Assign whole policies to training, development, and test groups before generating questions. Use NaviGator to draft student-style questions from training sections and review each pair for answerability, source match, and duplicates. Use development questions to select training, hybrid, and refusal settings. Development and test policies will remain searchable while their passages and questions stay out of fine-tuning pairs.

Target **150 held-out questions**, with about 120 answerable and 30 unanswerable from the frozen corpus. Use independently written or reviewed questions, including paraphrases and questions requiring multiple sections. Record the question ID, answerability, and accepted supporting section IDs. Freeze the labels before final scoring.

3. System architecture and model design

The diagram shows the architecture planned for the next deliverable. Its offline path prepares the corpus and trains the retriever. Its online path retrieves evidence and answers a student question.



Offline preparation and training planned for the next deliverable

The prototype I built for this project uses base BGE to encode policy passages and questions and ChromaDB to retrieve matching passages. For the controlled study, start from `BAAI/bge-small-en-v1.5` and fine-tune it with Sentence Transformers and PyTorch using a contrastive loss such as `MultipleNegativesRankingLoss`. The loss increases similarity between a question and its correct passage relative to unrelated passages. Avoid duplicate positives within a batch. A pilot will check data quality and training feasibility. Reviewed hard negatives are optional. Build distinct dense indexes for pretrained and fine-tuned embeddings and encode each query with its corresponding model.

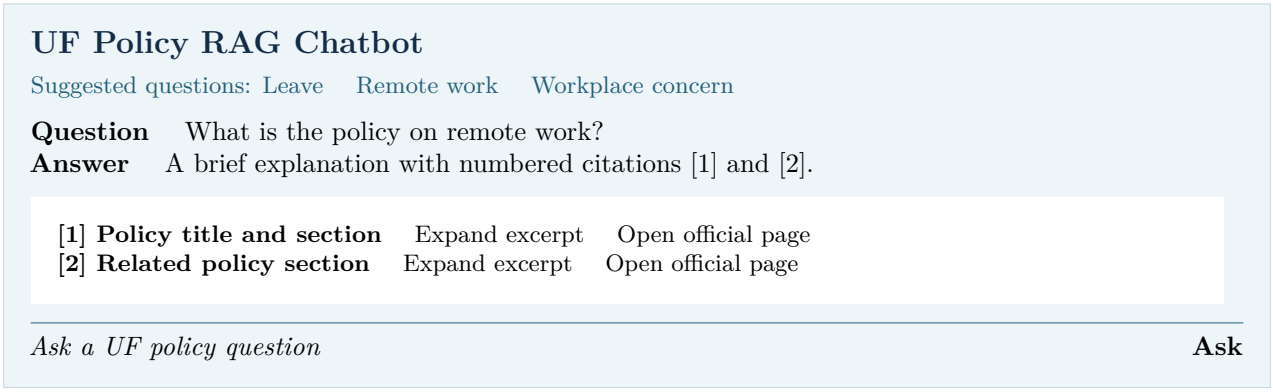
Four retrieval configurations planned for the next deliverable

Configuration	Purpose
BM25	Establish how well lexical matching retrieves relevant sections.
Pretrained BGE small	Establish the dense baseline from the starting checkpoint.
Fine-tuned BGE small	Isolate the effect of training on policy question–passage pairs.
BM25 + fine-tuned BGE	Test the effect of combining both rankings with reciprocal rank fusion.

In the online path, attach passage IDs and source metadata before sending context to the generator. Use the same corpus, section boundaries, top-five evidence budget, generation model, and prompt for the four conditions. Record the corpus version, model checkpoint, and retrieval settings for each run.

4. Streamlit interface design

I built a Streamlit chat that displays inline citations, retrieved excerpts, official links, and an adjustable passage count. The next deliverable will add retrieval-method selection and an explicit response when the corpus provides insufficient evidence.



Screen element	Design behavior
Question and suggestions	Accept plain language. Suggested questions demonstrate scope and help users get started.
Answer and citations	Display a short answer with inline numbers. Preserve applicable conditions and exceptions. Each number maps to a visible passage.
Source cards	Show policy title, section heading, excerpt, and an official link. Include the snapshot date where available.
Sidebar	Keep the adjustable passage count for interactive use. Add a retrieval-method selector in the next deliverable. Use five passages in scored comparisons.
Insufficient evidence	Add an explicit insufficient-evidence response in the next deliverable. Ask for clarification when ambiguity is the issue.
Loading or service error	Indicate that the request is in progress. If retrieval or generation fails, show a retry message and preserve the question.

An interface walkthrough will cover a supported answer, a question that needs clarification, a question outside the corpus, and an API failure. Each citation should open the matching evidence card, and each card should lead to the corresponding official page. Place a short notice near the input asking users to omit private details because the question and retrieved context are sent to the generation service.

5. Evaluation planned for the next deliverable

The central test is whether synthetic training questions help the retriever bridge informal student language and formal policy wording. Compare the pretrained and fine-tuned small checkpoints directly. Use the same question set across all four systems and report results overall and by vocabulary overlap. Define the overlap measure and cutoff using development data before opening the final test set.

Measure	Scoring rule
Recall@5	Fraction of annotated relevant sections covered by the five retrieved passages, averaged across answerable questions. Count a section once even if several passages match.
Mean reciprocal rank	Average the inverse rank of the first relevant passage on answerable questions. Use zero when no relevant passage is retrieved.
Answer correctness	Review whether substantive claims match the source and whether important conditions and exceptions are preserved.
Citation support	Divide the number of cited substantive claims supported by their cited passages by the total number of cited substantive claims. Record uncited substantive claims separately.
Appropriate refusal	On unanswerable items, count responses that acknowledge insufficient evidence. On answerable items, report incorrect refusals separately.

For questions requiring several pieces of evidence, record whether every required point has support. Use the same review rubric for all systems and record examples of wrong policy matches, missing exceptions, and citations that fail to support a claim. The completed 51-question BLEU, ROUGE, and METEOR comparison provides secondary answer-text diagnostics.

Risks and safeguards

Risk	Response
Synthetic examples are repetitive, incorrect, or too close to the final questions.	Split by policy first, review candidate pairs, remove near duplicates, and keep the final questions out of model selection.
Fine-tuning overfits or penalizes another valid source.	Pilot the small model, review negatives, annotate valid alternatives, and report results even when the trained model underperforms.
Documents change or API and compute access delay work.	Freeze a dated snapshot, log missing sources, verify access early, and prioritize the four core comparisons before optional hard-negative training.

Use public institutional documents for the corpus. Keep private user details out of research data. Report corpus coverage, the limits of manual review, and any unsupported answers. For consequential decisions, direct users to the official document or responsible UF office.

6. Implementation plan and submission

Completed during this project. I built the policy collection workflow and corpus snapshot, the pretrained BGE-base and ChromaDB baseline, the Streamlit chat with sourced answers, and the 51-question exploratory answer evaluation.

Deliverable 2 handoff. The blueprint documents the implementation. The notebook and repository items below still need attention before the GitHub submission.

Item	Completed	Pending
`notebooks/setup.ipynb`	Loads the prepared workbook, summarizes and previews records, and verifies ChromaDB.	Add collection provenance and date, missing-field and duplicate checks, a text-length plot, and a small live collection example. The current notebook loads the full prepared dataset.
GitHub repository	Project files and prototype were built for this course.	This workspace lacks Git metadata. Create a new repository, make the initial commit, push it, and submit the GitHub link alongside the PDF.

Next deliverable milestones

Sequence	Work	Completion evidence
1	Audit coverage and freeze the corpus.	Dated snapshot, source manifest, stable section IDs, and list of missing sources.
2	Split by policy and review synthetic training pairs.	Separate training and development sets plus an annotated held-out test set.
3	Implement BM25, run pretrained BGE-small, fine-tune BGE-small, and add reciprocal rank fusion.	Four retrieval configurations using the same corpus and passage budget.
4	Add retrieval-method selection and explicit insufficient-evidence behavior in Streamlit.	Walkthrough with a supported answer, an ambiguous question, and an unanswerable question.
5	Freeze settings and evaluate the held-out questions.	Recall@5, MRR, vocabulary-overlap results, answer correctness, citation support, and refusal results.

Design sources

- Rohan Bagulwar, *UF Policy RAG Chatbot — updated executive summary*, November 10, 2026, and the supplied project status notes.
- UF Regulation and Policy Hub
- UF NaviGator AI Toolkit
- BAAI BGE small model card and Sentence Transformers loss documentation